

ASAP - Status Update

Social Media API Feasibility Study - LinkedIn, X, and Bluesky

University of Toronto | ECE496 Capstone | September 2026

Supervisor: Prof. Hans-Arno Jacobsen | Research group: MSRG

Objective Evaluate official APIs for retrieving MSRG-owned account posts with low latency and minimal operating cost.

FEASIBILITY	LinkedIn	X (Twitter)	Bluesky
Official API	Community Management API	X API v2	AT Protocol / Public API
Mock account	ASAP Test Lab; app linked and Page association verified	@ASAPTestLasncf	asaptestlab.bsky.social
API access	Blocked: organization verification	Working; official API POC	Working; public-read POC
Text retrieval	Pending API approval	Tested	Tested public reads
Link retrieval	Pending API approval	Tested URL metadata	Tested external URI/title
Image retrieval	Pending API approval	Tested photo URL + dimensions	Tested CDN URLs + metadata
Incremental / polling	Not experimentally validated	since_id + empty polls tested; 5-minute interval planned	Expected feasible; incremental polling not validated
Authentication	OAuth + organization/Page authorization	App-Only Bearer Token	None for tested public reads
Rate / limits	Development Tier: 500 calls/app/day; 100/member/day	Console: 10,000 requests/15 min (app auth; tested timeline endpoint)	No practical concern identified for expected ASAP usage
Cost	No Standard Tier API fee identified in docs reviewed; production access, limits and costs need confirmation	Pay per returned resource; US\$8/month conservative budget estimate	\$0 for tested public reads
Overall status	Technically feasible; institutional access required	POC PASSED	PUBLIC-READ POC PASSED

KEY FINDINGS

- X** Text, links, images and incremental polling passed. Five-minute polling appears practical; billing depends on returned resources, not simply poll count.
- Bluesky** Tested public reads need no developer app, API key or organization verification; the simplest tested integration.
- LinkedIn** The blocker is institutional approval, not API failure. Production testing needs authorized MSRG/UoT identity and Page access.

Observed X billing: US\$5 prepaid; auto recharge off. Balance displayed about US\$4.98 after initial tests. Later incremental/empty polls showed no further visible decrease at console precision; this does not guarantee free empty polling.

NEXT STEPS

1. Obtain institutional LinkedIn access information and MSRG Page authorization.
2. Implement the unified provider interface and normalized post schema.
3. Connect working X and Bluesky providers to the ASAP mock backend/website.
4. Replace mock accounts with authorized MSRG accounts when access is available.

UNIFIED ARCHITECTURE / NEXT IMPLEMENTATION PHASE

